3 Real numbers and intervals

Sheet 13 Thu 19 Nov. 2026 *Sets of numbers: $\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$*

1. Warm-up · Quick Give the decimal form of $\frac{3}{8}$ and an approximation of $\frac{5}{6}$ to within 10^{-3}.
2. Core · Method Give the smallest set ($\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$) containing each of these numbers: -4; 0.25; $\frac{1}{3}$; $\sqrt{2}$; $\frac{6}{3}$.
3. Core + · True/false True or false? Justify. The number $\frac{1}{3}$ is a decimal number.
4. Review · Quick Calculate $(x+1)(x-3)$ for $x = 2$.
5. Thread · Quick How many faces, vertices and edges does a right triangular prism have?

Sheet 14 Mon 23 Nov. 2026 *Decimal bounds and rounding*

1. Warm-up · Quick Round 3.14159 to the nearest 10^{-2} and then to the nearest 10^{-3}.
2. Core · Method Bound $\sqrt{2}$ between two decimals that differ by 10^{-2}.
3. Core + · Code reading What is the final value of `k`? What can be deduced about $\sqrt{2}$?

```
k = 10
while (k/10)**2 < 2:
    k = k + 1
```

4. Review · Quick 5 kg of rice cost 12,500 Ar. How much do 8 kg cost?
5. Thread · True/false True or false? Justify. A quadrilateral whose diagonals have equal length is a rectangle.

Sheet 15 Tue 24 Nov. 2026 *Irrational numbers: $\frac{1}{3}$ is not decimal, $\sqrt{2}$ is irrational*

1. Warm-up · Quick Write $\sqrt{12}$ and $\sqrt{50}$ in the form $a\sqrt{b}$ with b the smallest possible integer.
2. Core · Sorting Among $0.333\ldots$; $\sqrt{9}$; $\sqrt{2}$; π; $\frac{22}{7}$, which are rational?
3. Core + · Reasoning Suppose $\sqrt{2} = \frac{p}{q}$ with $\frac{p}{q}$ in lowest terms, so $p^2 = 2q^2$. What can be said about the parity of p, then of q? What contradiction arises?
4. Review · Table Values $1\,;\,2\,;\,3\,;\,4$ with frequencies $5\,;\,8\,;\,4\,;\,3$. Give the total frequency, the relative frequency of the value 2 and the mean.

Value	1	2	3	4
Frequency	5	8	4	3

5. Thread · Quick Give the coordinates of the reflection of $A(3,-2)$ in the y-axis.

Sheet 16 Thu 26 Nov. 2026 *Intervals of $\mathbb{R}$: notation and representation*

1. Warm-up · Quick Compare -3 and -5, then $-\frac{1}{2}$ and -0.6.
2. Core · Method Write as an interval: $-2 \leqslant x < 5$; $x > 3$; $x \leqslant 0$.
3. Core + · MCQ The intersection $(-\infty, 2) \cap [-1, 4]$ is:

   A. $(2, 4]$ **B.** $(-\infty, 4]$ **C.** $[-1, 2)$ **D.** $[-1, 2]$
4. Review · Table Journey times (min).

Class	$[0,10)$	$[10,20)$	$[20,30)$
Frequency	4	9	7

What are the total frequency, the modal class and the relative frequency of $[10,20)$?

5. Thread · Problem One bus runs every 15 min, another every 20 min. They are together at 8 a.m. At what time are they next together?

Sheet 17 Mon 30 Nov. 2026 *Absolute value and distance between two real numbers*

1. Warm-up · Quick What is the distance between -3 and 5 on the number line?
2. Core · Quick Calculate $|-7|$, $|3-8|$ and $|\pi-4|$.
3. Core + · Method Solve $|x-3|=2$.
4. Review · Formula $f(x)=x^2-3$. Give the image of -2 and the preimages of 6.
5. Thread · Formula → table $f(x)=-x+3$. Complete: $f(-1)$, $f(0)$, $f(2)$.

Sheet 18 Tue 1st Dec. 2026 *Inequality $|x-a|\leqslant r$ and the associated interval*

1. Warm-up · Quick Solve $2x-3\leqslant 5$.
2. Core · Method Solve $|x-2|\leqslant 3$ and give S as an interval.
3. Core + · Translate Express as an inequality with absolute value: $x\in[-1,5]$ then $x\in(4,10)$.
4. Review · Problem Pack A: 6 yoghurts for €4.20. Pack B: 10 yoghurts for €6.80. Which is better value?
5. Thread · Quick A disc has radius 5 cm. Give its area and its circumference (exact values).